

	A	B	C	D	E	F	G	H	I
1	A sample of size 250 is drawn and mean is found to be 90. Test at 6% level of significance								
2	that whether it was drawn from population with mean 93 and SD 10 or not.								
3									
4	Solution :- Here we have								
5		Sample size(n) =	250						
6		Sample mean ($\bar{x}$) =	90						
7		Pop. Mean (μ) =	93						
8		Pop.SD(σ) =	10						
9	Here , we set up Hypothesis as								
10	$H_0 : \mu = 0$ i.e Sample is drawn from given hypothesis.								
11	$H_0 : \mu \neq 0$ i.e Sample is drawn from given hypothesis.								
12									
13	Under H_0 ,test statistics								
14		S.E($\bar{x}$) =	0.632456	= D8/SQRT(D5)					
15		Z_{cal} =	-4.74342	= (D6 - D7) /D14					
16		Z_{cal} =	4.743416	=ABS(D15)					
17	Level of significance (α) = 0.06								
18		Z_{tab} =	1.880794	= NORMSINV(1 - D17/2)					
19	Decision: - Since $Z_{cal} > Z_{tab}$,we accept H_1 and reject H_0 with the								
20	conclusion that sample is drawn from different population.								
21									
22									
23	Name : Hari Bashyal								
24	Roll no. : 80010538								

	A	B	C	D	E	F	G	H	I
1	From the following information , state whether Company A is superior to								
2	Company B or not.								
3		For company A		For company B					
4		$n_1 =$	100	$n_1 =$	125				
5		$x^*_1 =$	260	$x^*_2 =$	258				
6		$s_1 =$	5	$s_2 =$	10				
7									
8	Solution: - Here we have								
9		For company A		For company B					
10		$n_1 =$	100	$n_1 =$	125				
11		$x^*_1 =$	260	$x^*_2 =$	258				
12		$s_1 =$	5	$s_2 =$	10				
13		Here , we set up Hypothesis as							
14		$H_0 : \mu_1 = \mu_2$ i.e Sample is drawn from given hypothesis.							
15		$H_0 : \mu_1 > \mu_2$ i.e Sample is drawn from given hypothesis.							
16									
17		Under H_0 ,test statistics							
18		$S.E(x^*_1 - x^*_2) =$			1.025	$= \text{SQRT}(C12^2/C10+E12^2/E10)$			
19				$Z_{cal} =$	1.9518	$= (C11-E11)/E18$			
20		Level of significance (α) =			0.05				
21		For OTT,	$Z_{tab} =$	1.645	$= \text{NORMSINV}(1 - E20)$				
22		Decision: - Since $Z_{cal} > Z_{tab}$,we reject H_0 and accept H_1 with the							
23		conclusion that Company A is superior to Company B							
24									
25									
26		Name : Hari Bashyal							
27		Roll no. : 80010538							